THE UNIVERSITY OF AZAD JAMMU & KASHMIR MUZAFFARABAD



Name: Zoya Mazhar

Roll no: 2024-SE-26

Course: Software Consyruction

Submitted to: Mam Asma

Department of Software Engineering

Q1: Online Ticket Reservation System

Objective

To develop a use case diagram for an online ticket reservation system and identify its actors and use cases.

Introduction

An online ticket reservation system allows customers to search for available tickets, select tickets, make reservations, and pay online. It also allows the administrator to manage tickets and reservations.

Actors

Customer: Searches for tickets, selects tickets, makes reservations, pays online, and receives

Administrator: Manages ticket information and reservations.

Payment Gateway: Processes online payments and returns the payment status.

Use Cases

Register / Log In

Search Tickets

View Availability

Select Ticket

Book Ticket

Make Payment

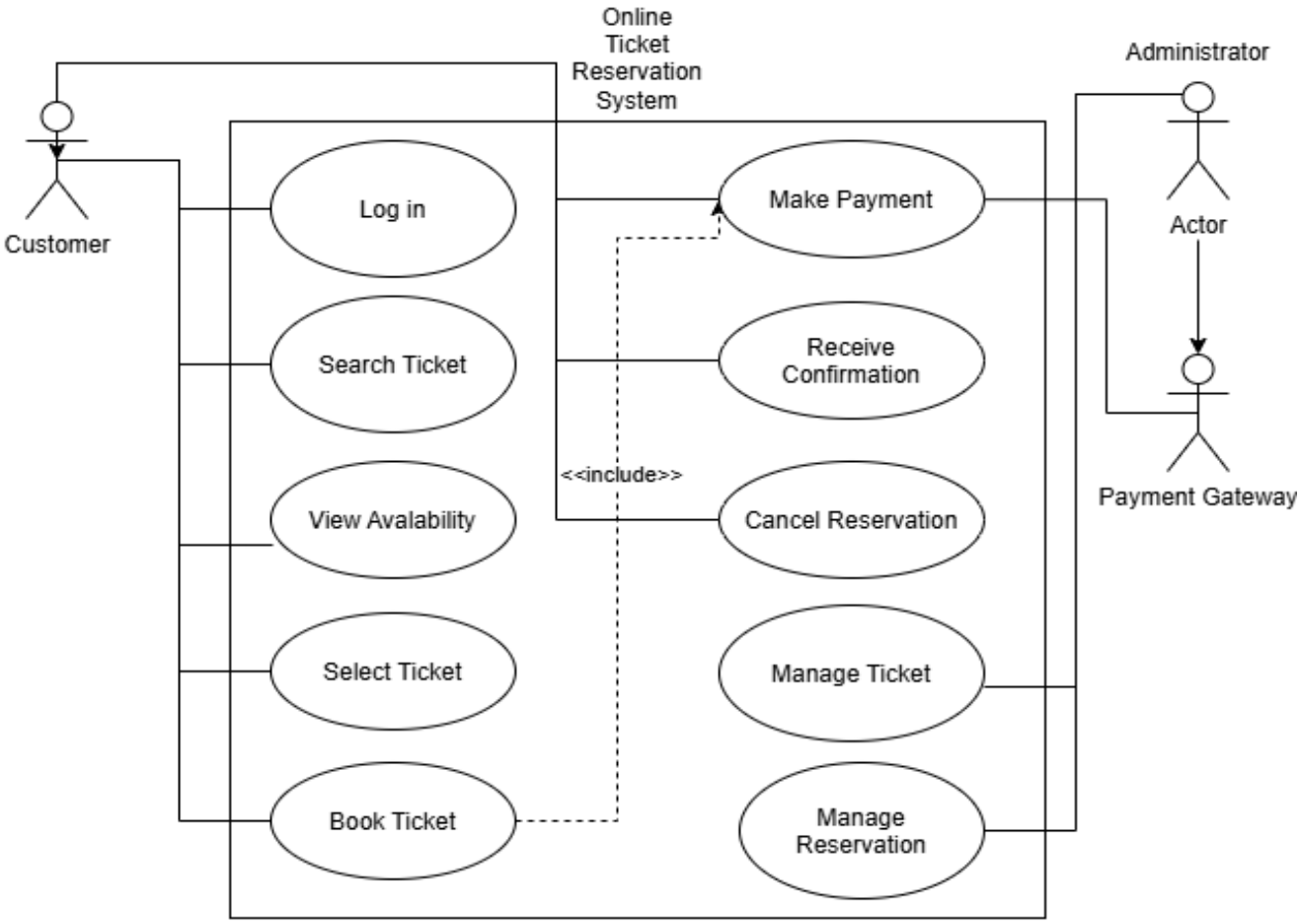
Receive Confirmation

Cancel Reservation

Manage Tickets

Manage Reservations

Use Case Diagram



Conclusion

The use case diagram shows the main functions of an online ticket reservation system and how the customer, administrator, and payment gateway interact with it.

Q2: Credit Card Processing System

Objective

To identify the actors and use cases of a credit card processing system and develop its use case diagram.

Introduction

A credit card processing system is used when a customer purchases an item from a shop and pays using a credit card. The vendor prepares the bill, processes the card transaction, and provides a receipt. The customer signs the receipt and returns it to the vendor.

Actors

Customer: Selects an item, provides the credit card, signs the receipt, and returns the signed receipt.

Vendor: Prepares the bill, operates the card reader, enters the transaction amount, and provides the receipt.

Bank / Credit Card Authorization System: Verifies the credit card, authorizes the transaction, and supports the updating of the credit balance after an approved transaction.

Use Cases

Select Item

Generate Bill

Swipe Credit Card

Enter Transaction Amount

Verify Credit Card

Process Transaction

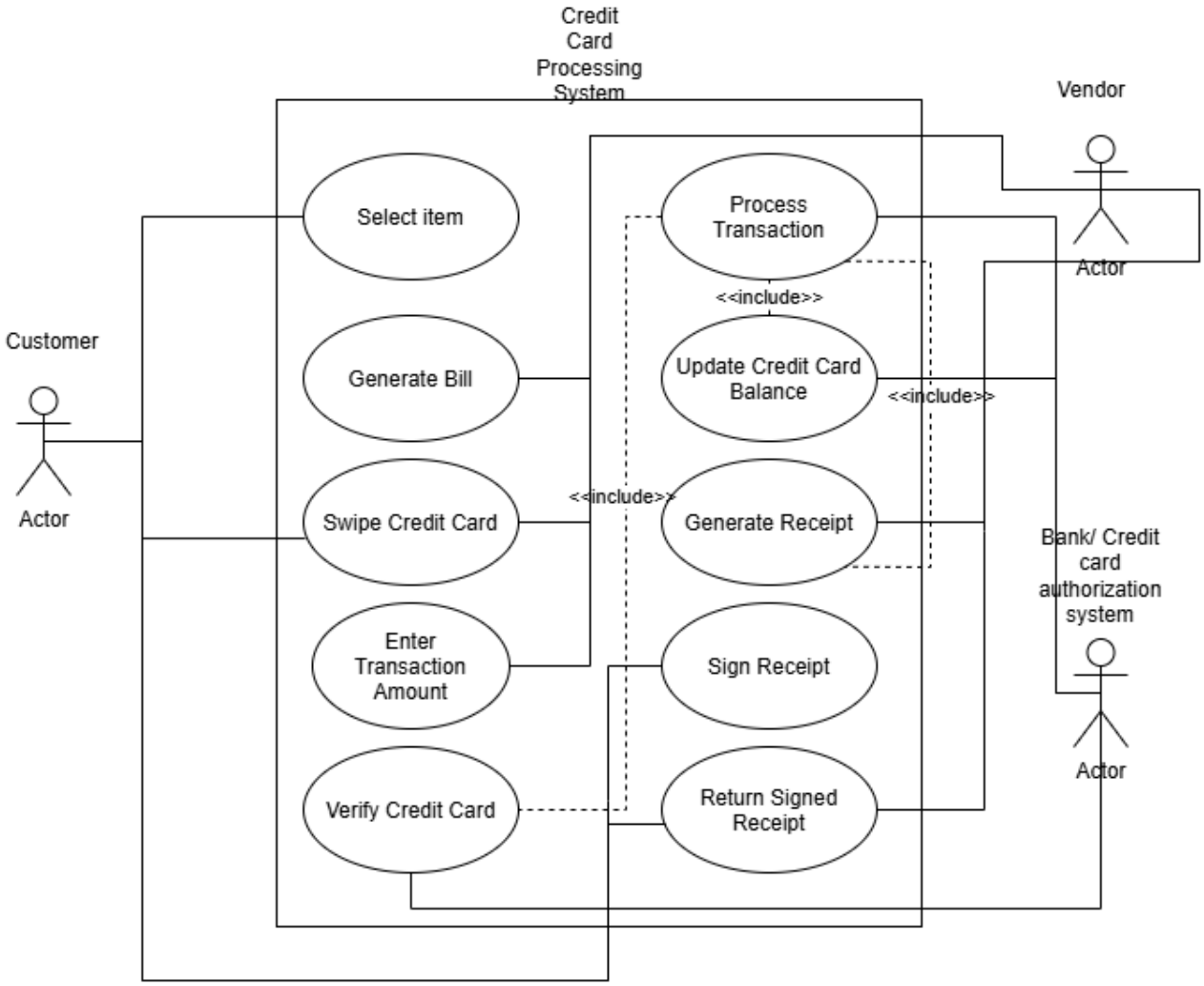
Update Credit Card Balance

Generate Receipt

Sign Receipt

Return Signed Receipt

Use Case Diagram



Conclusion

The use case diagram represents the main activities involved in credit card processing. It shows how the customer, vendor, and bank or credit card authorization system participate in the transaction.